# Random Forest Classification

2026-06-14

## Contents


**1 Crime Type Classification** **2**
1.1 Hyperparameter Tuning . . . 2
1.2 Threshold Tuning . . . 3
1.3 Model Performance . . . 4
1.4 Variable Importance . . . 5

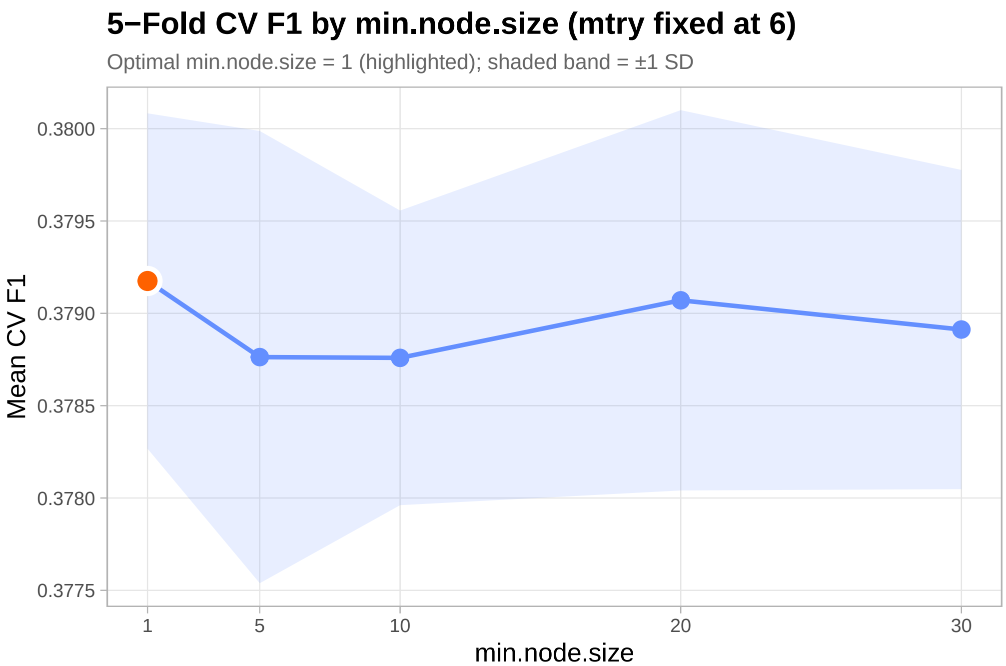
**2 Crime Subtype Classification** **6**
2.1 Hyperparameter Tuning . . . 6
2.2 Final Model . . . 6
2.3 Performance . . . 7

# 1 Crime Type Classification

## 1.1 Hyperparameter Tuning

Table 1: 5-Fold CV F1 by min.node.size (mtry fixed at 6)

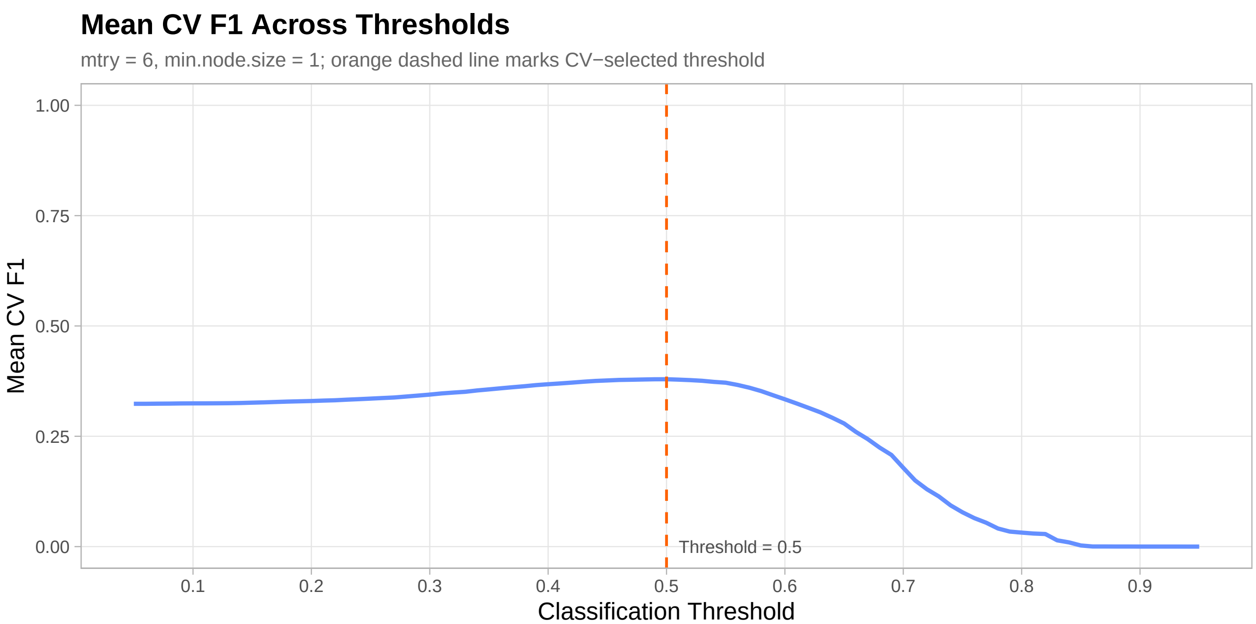
| min.node.size | Mean CV F1 | SD | Selected |
|---|---|---|---|
| 1 | 0.3792 | 0.0009 | Yes |
| 5 | 0.3788 | 0.0012 | |
| 10 | 0.3788 | 0.0008 | |
| 20 | 0.3791 | 0.0010 | |
| 30 | 0.3789 | 0.0009 | |



## 1.2 Threshold Tuning

Table 2: 5-Fold CV F1 at Selected Threshold (0.5)

| Fold | F1 |
|---|---|
| 1 | 0.3787 |
| 2 | 0.3801 |
| 3 | 0.3784 |
| 4 | 0.3782 |
| 5 | 0.3809 |
| Mean | 0.3793 |



### 1.3 Model Performance

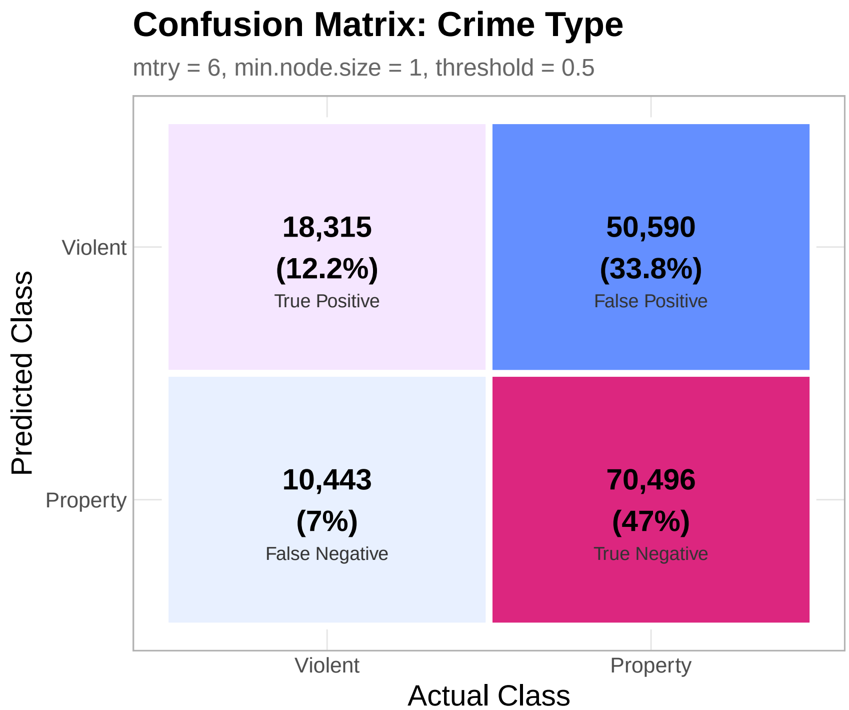
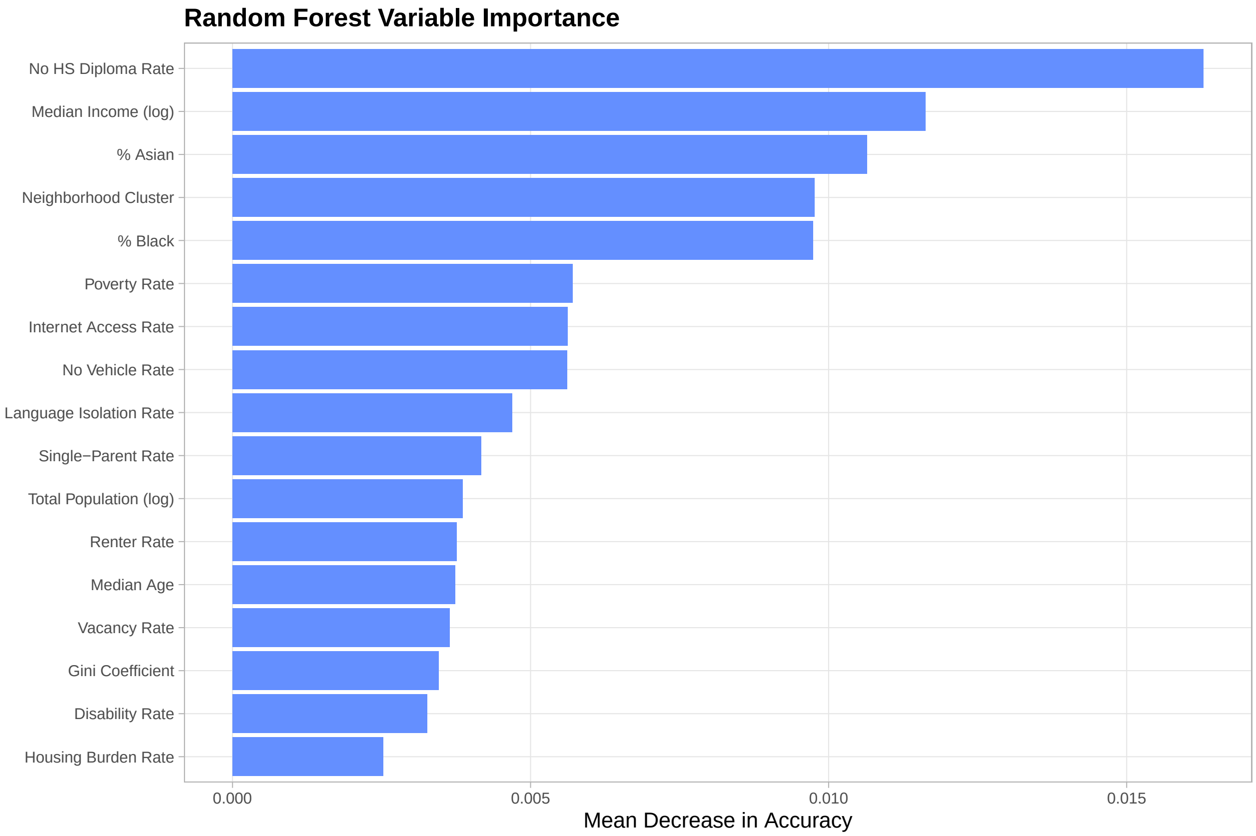



Table 3: Random Forest Classification Performance

| Metric | Value |
|---|---|
| F1 Score | 0.3751 |
| Matthews Correlation Coefficient | 0.1731 |
| Accuracy | 0.5927 |
| Sensitivity / Recall | 0.6369 |
| Specificity | 0.5822 |
| Precision | 0.2658 |
| Balanced Accuracy | 0.6095 |

## 1.4 Variable Importance

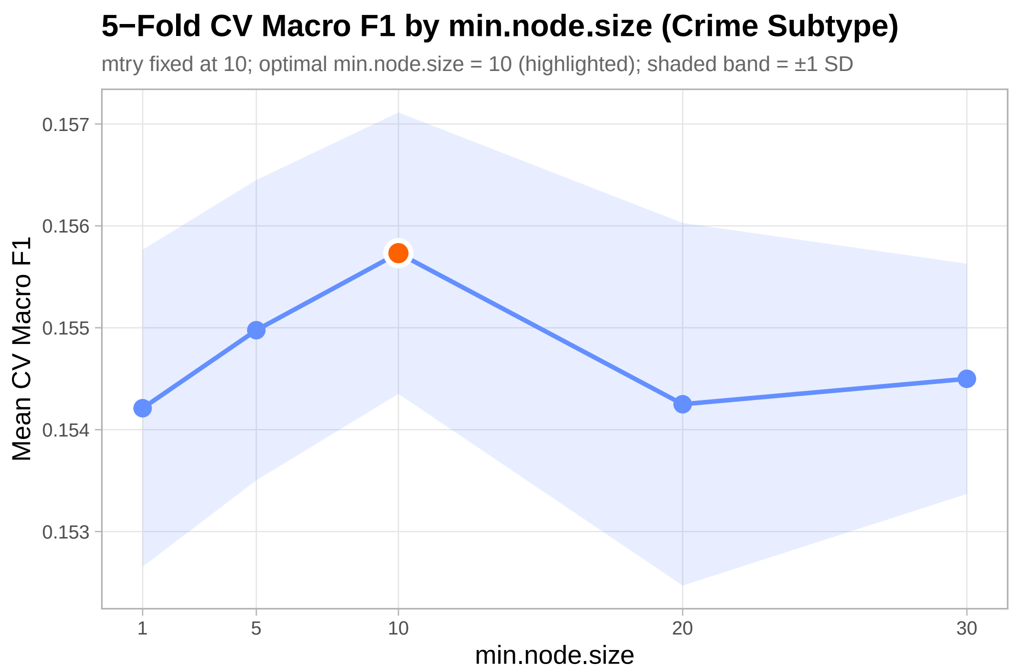


# 2 Crime Subtype Classification

## 2.1 Hyperparameter Tuning

Table 4: 5-Fold CV Macro F1 by min.node.size (mtry fixed at 10)

| min.node.size | Mean CV Macro F1 | SD | Selected |
|---|---|---|---|
| 1 | 0.1542 | 0.0016 | |
| 5 | 0.1550 | 0.0015 | |
| 10 | 0.1557 | 0.0014 | Yes |
| 20 | 0.1542 | 0.0018 | |
| 30 | 0.1545 | 0.0011 | |




## 2.2 Final Model

```
## Growing trees.. Progress: 7%. Estimated remaining time: 8 minutes, 0 seconds.
## Growing trees.. Progress: 14%. Estimated remaining time: 6 minutes, 58 seconds.
## Growing trees.. Progress: 21%. Estimated remaining time: 6 minutes, 19 seconds.
## Growing trees.. Progress: 28%. Estimated remaining time: 5 minutes, 38 seconds.
## Growing trees.. Progress: 36%. Estimated remaining time: 4 minutes, 59 seconds.
## Growing trees.. Progress: 43%. Estimated remaining time: 4 minutes, 26 seconds.
## Growing trees.. Progress: 50%. Estimated remaining time: 3 minutes, 55 seconds.
## Growing trees.. Progress: 57%. Estimated remaining time: 3 minutes, 21 seconds.
## Growing trees.. Progress: 64%. Estimated remaining time: 2 minutes, 47 seconds.
## Growing trees.. Progress: 71%. Estimated remaining time: 2 minutes, 15 seconds.
## Growing trees.. Progress: 78%. Estimated remaining time: 1 minute, 42 seconds.
## Growing trees.. Progress: 84%. Estimated remaining time: 1 minute, 12 seconds.
## Growing trees.. Progress: 91%. Estimated remaining time: 39 seconds.
## Growing trees.. Progress: 98%. Estimated remaining time: 7 seconds.
## Computing permutation importance.. Progress: 18%. Estimated remaining time: 2 minutes, 21 seconds.
## Computing permutation importance.. Progress: 35%. Estimated remaining time: 1 minute, 53 seconds.
## Computing permutation importance.. Progress: 53%. Estimated remaining time: 1 minute, 23 seconds.
```

```
## Computing permutation importance.. Progress: 70%. Estimated remaining time: 53 seconds.
## Computing permutation importance.. Progress: 88%. Estimated remaining time: 20 seconds.
```

## 2.3 Performance

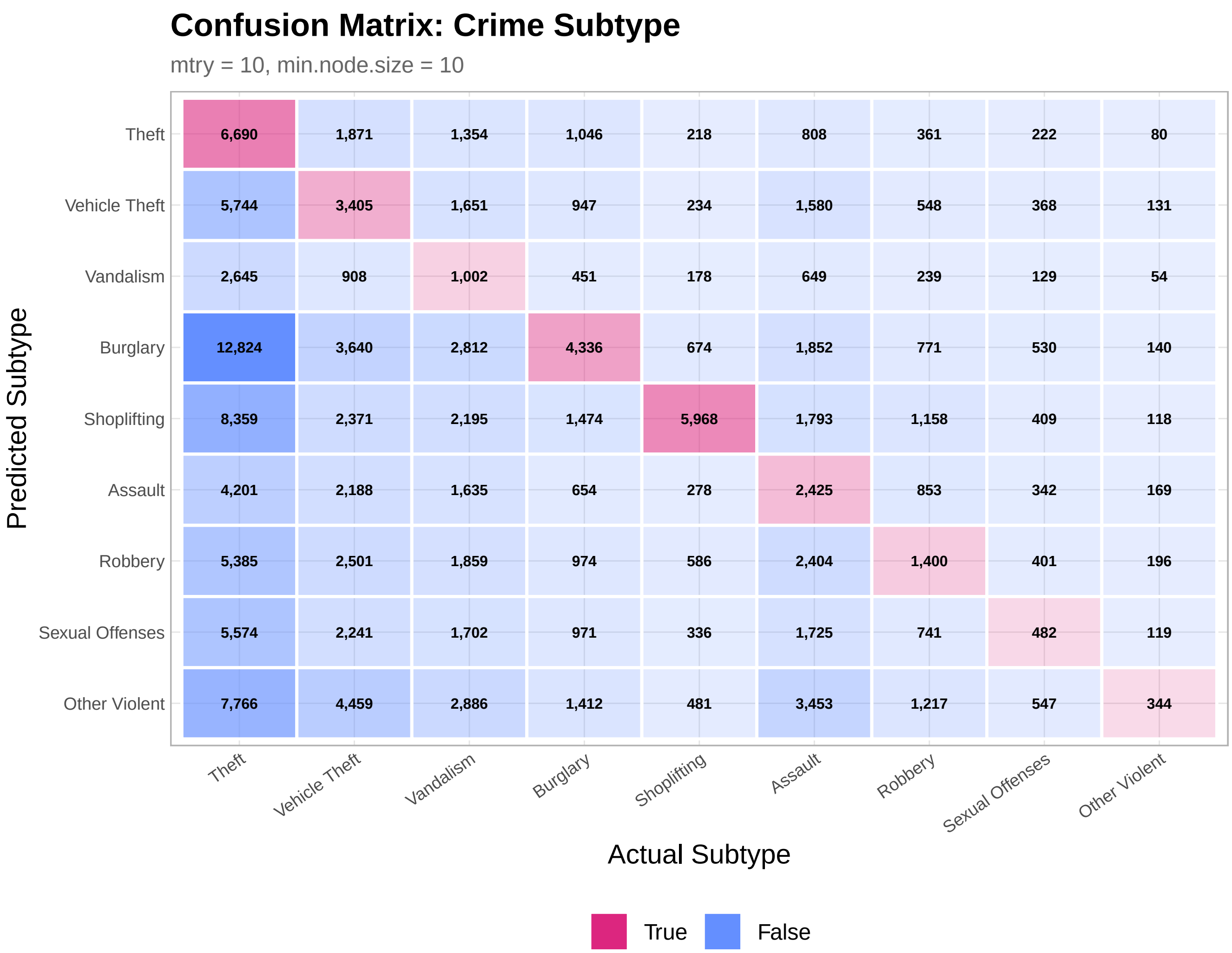



Table 5: Per-Subtype F1 Score

| Subtype | Type | F1 |
|---|---|---|
| **Property** | | |
| Theft | Property | 0.186 |
| Vehicle Theft | Property | 0.178 |
| Vandalism | Property | 0.086 |
| Burglary | Property | 0.218 |
| Shoplifting | Property | 0.364 |
| **Violent** | | |
| Assault | Violent | 0.165 |
| Robbery | Violent | 0.122 |
| Sexual Offenses | Violent | 0.056 |
| Other Violent | Violent | 0.029 |

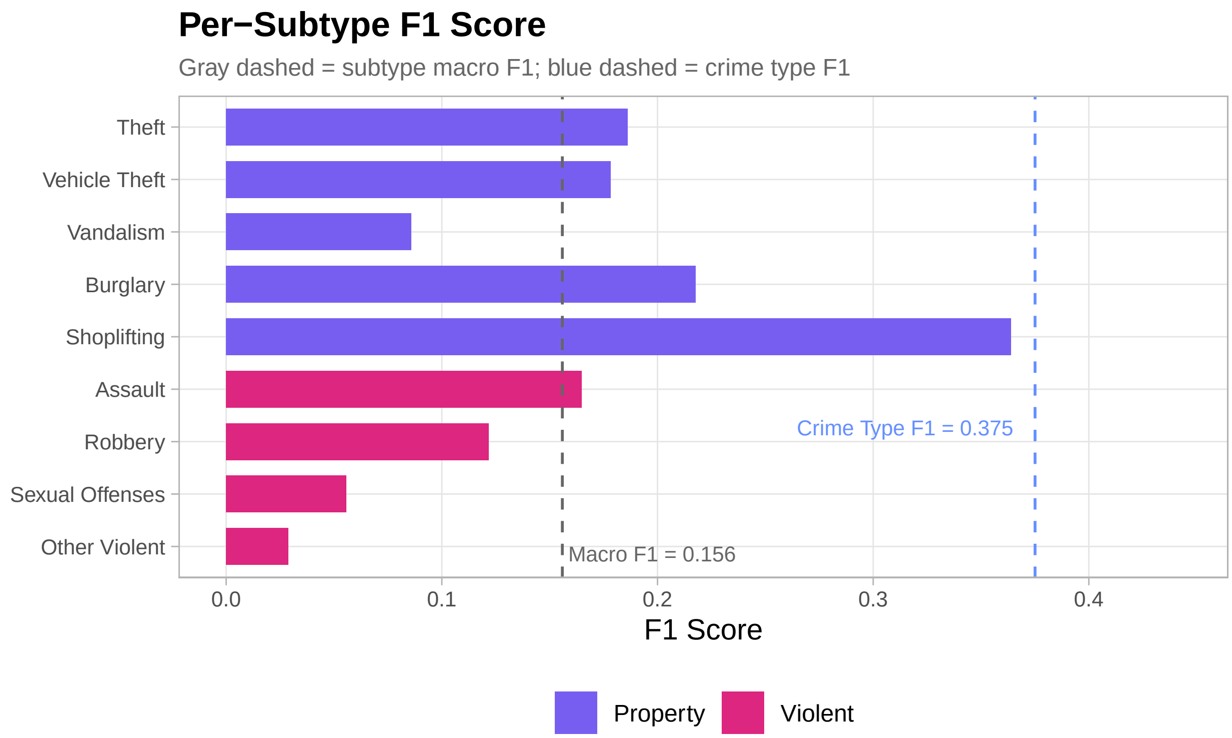
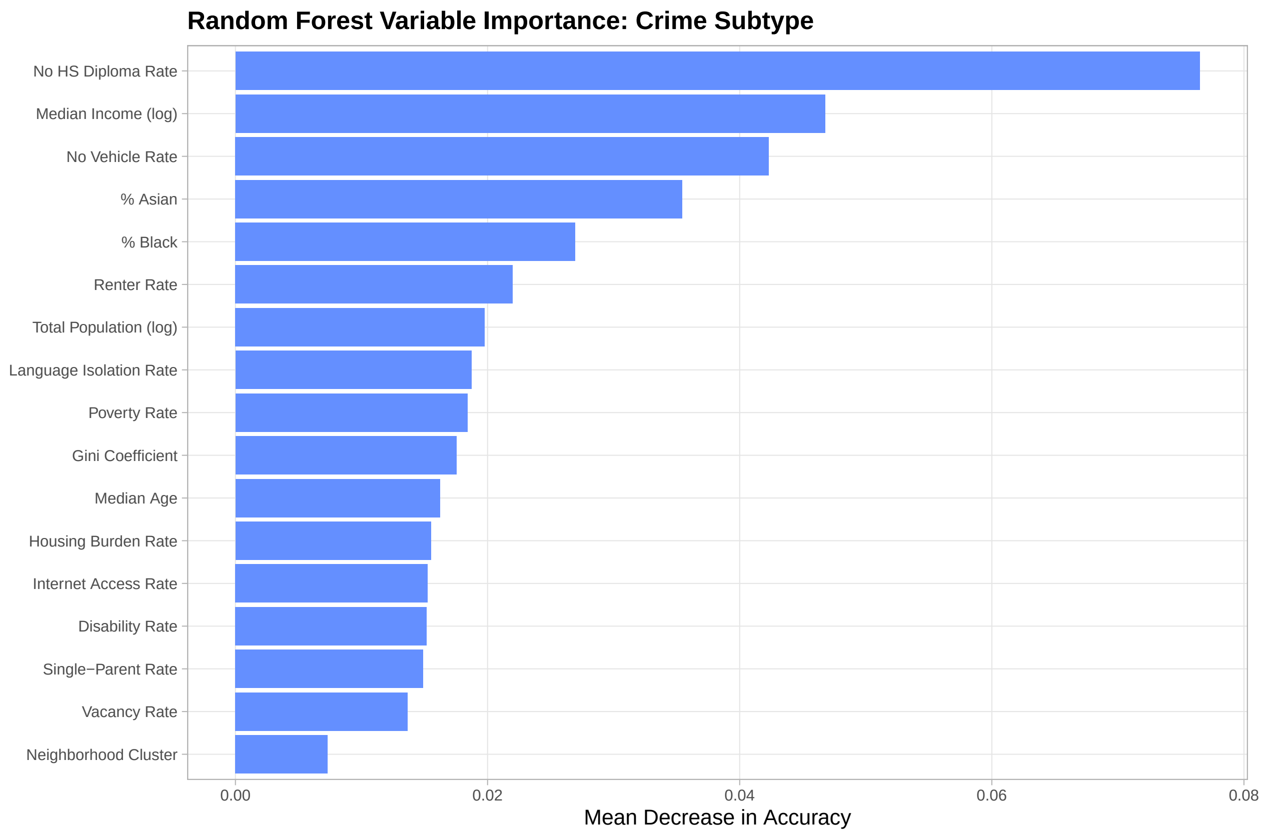



Table 6: Overall Crime Subtype Performance (Test Set)

| Metric | Value |
|---|---|
| Macro F1 | 0.1559 |
| MCC | 0.0942 |
| Weighted F1 | 0.1768 |
| Overall Accuracy | 0.1739 |


Random Forest Variable Importance: Crime Subtype
No HS Diploma Rate
Median Income (log)
No Vehicle Rate
% Asian
% Black
Renter Rate
Total Population (log)
Language Isolation Rate
Poverty Rate
Gini Coefficient
Median Age
Housing Burden Rate
Internet Access Rate
Disability Rate
Single−Parent Rate
Vacancy Rate
Neighborhood Cluster
0.00
0.02
0.04
0.06
0.08
Mean Decrease in Accuracy